THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 1

Chapter 3: **The Atomic Foundations of Matter** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The law of conservation of mass was proposed by: 1

(a) Antoine Lavoisier **(b)** J. J. Thomson
(c) Joseph Proust **(d)** John Dalton

2. The atomicity of phosphorus (P_4) is: 1

(a) 1 **(b)** 4 **(c)** 8 **(d)** 2

3. An anion is a: 1

(a) molecule **(b)** positively charged ion
(c) negatively charged ion **(d)** neutral particle

4. The law of constant proportions was proposed by: 1

(a) Rutherford **(b)** Joseph Proust **(c)** Dalton **(d)** Lavoisier

5. The atomicity of sulphur (S_8) is: 1

(a) 4 **(b)** 1 **(c)** 2 **(d)** 8

6. The chemical formula of sodium chloride is: 1

(a) Na_2Cl_2 **(b)** $NaCl$ **(c)** $NaCl_2$ **(d)** Na_2Cl

7. According to the law of conservation of mass, in a chemical reaction the total mass of the reactants: 1

(a) is less than the products **(b)** equals the total mass of the products
(c) is zero **(d)** is greater than the products

8. The molecular mass of H_2O is (H = 1, O = 16): 1

(a) 16 u **(b)** 18 u **(c)** 20 u **(d)** 17 u

9. The formula of calcium chloride is: 1

(a) $CaCl$ **(b)** Ca_2Cl_2 **(c)** Ca_2Cl **(d)** $CaCl_2$

10. 12 g of carbon burns completely in 32 g of oxygen. The mass of carbon dioxide formed is: 1

(a) 28 g **(b)** 12 g **(c)** 44 g **(d)** 20 g

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. State the law of conservation of mass. Who proposed it? 2

12. What is atomicity? Give the atomicity of H_2, O_3 and P_4. 2

13. How is a covalent bond formed? Give one example. 2

14. State the law of constant proportions with an example. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A reaction between 6.0 g of carbon and 16.0 g of oxygen gives CO2. Verify the law of conservation of mass. 3

16. State three properties of ionic compounds. 3

17. Differentiate between an atom and a molecule (three points). 3

OR

In a compound, hydrogen and oxygen are in the ratio 1 : 8 by mass. Find the mass of oxygen in 27 g of water. 3

18. State three properties of covalent compounds. 3

OR

Calculate the mass of oxygen needed to burn completely 3 g of carbon to form CO2 (C : O = 12 : 32). 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe an experiment to verify the law of conservation of mass using sodium sulphate and barium chloride solutions in a closed flask. State the observation and conclusion. 5

OR

A compound contains carbon and oxygen in the ratio 3 : 8. Find the mass of the compound formed when 18 g carbon is burnt and the mass of oxygen needed. Name the compound. 5

20. State Dalton's atomic theory with five postulates and mention which points were modified later. 5

OR

Calculate the molecular masses of H_2O, CH_4, C_2H_5OH, HNO_3 and Na_2CO_3 (H = 1, C = 12, N = 14, O = 16, Na = 23). 5

END OF QUESTION PAPER — **Total: 40 Marks**

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